

ABSALOM RANELLETTI

✉ absalom.ranelletti@gmail.com [in LinkedIn](#) [Github](#) ☎ 805-433-3000 [Portfolio](#)

Simi Valley, California

Summary

I am a soon-to-be-graduate of UC Santa Cruz looking for an entry level IT position where I can contribute to the maintenance of technical systems, troubleshooting at the company level, and implementation of new systems. I have the theoretical underpinning of IT architecture to quickly grasp how to maintain various systems.

Education

University of California, Santa Cruz (UCSC), Santa Cruz, CA

Graduating June 2026

B.A. in Computer Science 3.64 GPA

Skills

Languages: Python, Java, C, C++, JavaScript

Software: Microsoft Office, Excel/Google Sheets, Google Products, Wireshark, Mininet

General: Research, Communication, Technical Writing, Science Communication, Cybersecurity, AI management, Machine Learning, OSI protocol stack, Computer Networking Principles, Project Leadership

Frontend: HTML, CSS, LaTeX

Developer Tools: VS Code, GitHub, Google Scripts, GDB, Shell scripting, Openflow/POX

Quickly learns new computer programs, programming languages, and software systems.

Projects

Links to the code for each project (and more) can be found on my [portfolio site](#).

Chat Program

Winter 2026

- Developed a peer-to-peer chat application coordinated by a server using Python socket programming with TCP.
- Coordinated development as project lead, integrating client and server components and designing an application-layer messaging protocol, elaborating on provided project specs, to ensure consistent communication.

Machine Learning (AI) with Synthetic Data (Python)

Winter 2025

- Cleaned and analyzed synthetic datasets in Python; trained and evaluated multiple machine learning models (logistic regression, k-NN, decision tree) using scikit-learn, achieving 96% mean accuracy.
- Performed statistical performance analysis and documented technical decisions and results in a formal written report.

Chess Engine and AI (C++)

Fall 2025

- Developed a C++ chess engine with an ImGui-based interface, implementing valid move logic and integrating instructor-provided and third-party code into an existing codebase to meet project specifications.
- Implemented and optimized a negamax-based AI using bitboards and alpha-beta pruning; troubleshot integration issues by analyzing dependencies, debugging complex errors, and clarifying requirements through effective technical communication.

Multi-Threaded HTTPS Server (C)

Fall 2025

- Wrote and implemented a multi-threaded HTTPS server in C supporting concurrent GET and PUT requests; ensured data integrity using custom bounded queue and reader-writer locks, handling hundreds of requests in seconds.
- Applied systems design principles (layering, modularity, abstraction) and strengthened debugging skills by diagnosing concurrency and tooling issues in a complex, low-level environment.

Work Experience

Storage Scholars

March 2025 – June 2025

Student Business Development Representative

Remote

- Improved sales tactics by researching UCSC regulations, adjusting the marketing plan to comply, then writing a web-scraping and emailing script.
- Gained familiarity with Slack and TalentLMS.

Ventura County Elections Division

March 2020

Poll Worker

- Set up a polling station with a team, managed the voter roll and issued provisional ballots on election day.